

5(a)	<i>recessive</i> phenotype of (recessive) allele is masked by a dominant allele or two copies of the (recessive) allele are needed for phenotype (to be displayed) or alleles only affect the phenotype in the absence of a dominant allele or (genotype) must be homozygous for the (recessive) allele to affect the phenotype ; <i>codominant</i> both alleles affect the phenotype (in the heterozygote) ;	2										
5(b)	<table><tr><th>blood group</th><th>genotypes</th></tr><tr><td>A</td><td>$I^A I^A$ $I^A I^O$;</td></tr><tr><td>B</td><td>$I^B I^B$ $I^B I^O$;</td></tr><tr><td>AB</td><td>$I^A I^B$</td></tr><tr><td>O</td><td>$I^O I^O$;</td></tr></table>	blood group	genotypes	A	$I^A I^A$ $I^A I^O$;	B	$I^B I^B$ $I^B I^O$;	AB	$I^A I^B$	O	$I^O I^O$;	3
blood group	genotypes											
A	$I^A I^A$ $I^A I^O$;											
B	$I^B I^B$ $I^B I^O$;											
AB	$I^A I^B$											
O	$I^O I^O$;											

Question	Answer	Marks															
5(c)	parental phenotypes: (B Rh+ O Rh+) parental genotypes: I^B I^O Dd I^O I^O Dd <table><tr><td>parental gametes</td><td>I^BD</td><td>I^Bd</td><td>I^OD</td><td>I^Od</td></tr><tr><td>I^OD</td><td>I^BI^ODD B Rh+</td><td>I^BI^ODd B Rh+</td><td>I^OI^ODD O Rh+</td><td>I^OI^ODd O Rh+</td></tr><tr><td>I^Od</td><td>I^BI^ODd B Rh+</td><td>I^BI^Odd B Rh-</td><td>I^OI^ODd O Rh+</td><td>I^OI^Odd O Rh-</td></tr></table> ratio of offspring phenotypes: B Rh+ O Rh+ B Rh- O Rh- 3 : 3 : 1 : 1 allow one mark for correct parental genotypes ; allow one mark for correct parental gametes ; allow one mark for correct offspring genotypes ; allow one mark for correct, matching, phenotypes ; allow one mark for correct ratio ; apply ecf – if incorrect parental phenotype for gene I, ecf for Rhesus gene, evidence of correct parental gametes, offspring genotypes and matching phenotypes to max 3 marks	parental gametes	I ^B D	I ^B d	I ^O D	I ^O d	I ^O D	I ^B I ^O DD B Rh+	I ^B I ^O Dd B Rh+	I ^O I ^O DD O Rh+	I ^O I ^O Dd O Rh+	I ^O d	I ^B I ^O Dd B Rh+	I ^B I ^O dd B Rh-	I ^O I ^O Dd O Rh+	I ^O I ^O dd O Rh-	5
parental gametes	I ^B D	I ^B d	I ^O D	I ^O d													
I ^O D	I ^B I ^O DD B Rh+	I ^B I ^O Dd B Rh+	I ^O I ^O DD O Rh+	I ^O I ^O Dd O Rh+													
I ^O d	I ^B I ^O Dd B Rh+	I ^B I ^O dd B Rh-	I ^O I ^O Dd O Rh+	I ^O I ^O dd O Rh-													